

Oligopoly

Assumptions of an Oligopoly Market

An oligopoly market rests on a different set of assumptions than the perfectly competitive market I analyzed earlier in the course. First, only a small number of large firms supply the market — the Juice Mogul oligopoly simulation placed just twelve firms in direct competition, the same twelve firms that were modeled as price takers in the perfect-competition version. Second, significant barriers to entry protect those firms' positions, so the field of competitors stays fixed rather than expanding freely whenever profit opportunities appear. Third, and most importantly, firms in an oligopoly are mutually interdependent: because so few sellers exist, each firm's pricing decision visibly affects its rivals' sales, and every rival is assumed to notice and react. This interdependence is what separates oligopoly from both perfect competition and monopolistic competition — a firm cannot set price or output in isolation the way a monopolist can, but it is also not a pure price taker the way a perfectly competitive firm is. Finally, because interdependence creates a shared interest in higher prices, oligopolists face a constant temptation to collude and act as a single monopolist, forming a cartel that restricts output and raises price above the competitive level — but every member also has an individual incentive to secretly undercut the agreed price and capture a larger share of the market, which is precisely the prisoner's-dilemma dynamic that played out repeatedly in the simulation (Shapiro et al., 2022).

Game Strategy and Results

My strategy throughout the ten rounds was to undercut my rivals' prices rather than hold a fixed price or wait passively for a cartel to form around me. My participation in cartels was inconsistent: in some rounds other firms invited me into a price-fixing agreement and I held the higher, coordinated price alongside the group, while in other rounds I was left out — either because the group had already settled without me or because the agreement broke down — and I undercut whatever price the cartel had set. This mix of cooperating when possible and undercutting when not produced a cumulative profit of \$71,174.00, placing me fifth out of the twelve competing firms. My result sat in a tight cluster just below Fruit & Veggie Bar (\$71,716.00) and comfortably ahead of Vitafusion in sixth place (\$69,507.00), while the top three firms — The Juice Press (\$80,207.00), Just Squeezed (\$79,715.00), and Oasis Juice Bar (\$75,123.00) — pulled meaningfully ahead of the rest of the field (see Appendix), suggesting they sustained cooperative pricing across more of the ten rounds than I did.

Is This Market Efficient?

This market was less efficient than the perfectly competitive version of the same simulation. In perfect competition, free entry and rivalry pushed every firm's profit down to a normal \$2,500, the textbook signature of allocative efficiency (Shapiro et al., 2022). Here, every one of the twelve firms earned well above that figure — profits ranged from \$58,632.00 to \$80,207.00 — which indicates that prices stayed above marginal cost for most of the game rather than being competed down to a normal-profit level. That gap follows directly from the assumptions above: with only twelve interdependent sellers and real barriers to entry, no single firm's price cutting was ever enough to force the whole market back toward the competitive outcome the way it would with many small, powerless firms. At the same time, the market was not perfectly inefficient either. The instability of the cartels I was sometimes part of, and the wide \$21,575 spread between the top and bottom of the leaderboard, shows that collusion never fully hardened into a single monopoly price. Every firm's temptation to undercut the group, mine included, kept pulling prices back down toward a more competitive level, so the market settled somewhere between the monopoly outcome and the perfectly competitive one — exactly where oligopoly theory predicts it should land.

What I Learned

The biggest lesson from this simulation was that success in an oligopoly depends less on any single pricing decision and more on managing the tension between cooperation and defection round after round. Undercutting felt like the safer, more controllable strategy in the moment — it guaranteed sales even when I could not be sure a cartel would hold — but the firms that finished ahead of me appeared to sustain cooperative pricing more consistently, which earned them a steadier and ultimately larger cumulative profit. I also saw how directly this connects to the game theory concepts from earlier in the course: every round inside a cartel was effectively a repeated prisoner's dilemma, where the dominant individual-round strategy (undercut) was not the same as the strategy that maximized joint, and it turned out personal, long-run profit (cooperate).

What I Would Do Differently

If I played again, I would commit more consistently to whatever cartel price was on the table instead of defecting whenever undercutting looked profitable in a single round, since the top three firms' results suggest sustained cooperation outperformed my mixed strategy over ten rounds. I would also pay closer attention to how many firms were actually part of a given cartel before deciding whether to trust it. Smaller cartels were noticeably easier to sustain in the simulation — with fewer firms, it was easier to notice who had broken the agreed price and to informally punish or exclude them the following round, and each member had more to lose individually from starting a price war. As the number of participating firms grew, holding the cartel together got harder: more members meant more chances for any single firm to defect unnoticed, more incentive for each one to grab extra volume before someone else did, and a larger cartel started to behave less like a single coordinated monopolist and more like a loosely competitive market. That pattern matched my own experience of being welcomed into some cartels and shut out of or defected against in others — the larger or looser the group, the less reliably it held.

References

Shapiro, D., MacDonald, D., & Greenlaw, S. A. (2022). Principles of microeconomics 3e. OpenStax.
<https://openstax.org/books/principles-microeconomics-3e/pages/10-2-oligopoly>

Appendix

Figure 1

Final Leaderboard, Juice Mogul: Oligopoly Simulation

JUICE MOGUL		Oligopoly Game Over
Congratulations! Final Score		
Name	Profit	
The Juice Press	\$80,207.00	
Just Squeezed	\$79,715.00	
Oasis Juice Bar	\$75,123.00	
Fruit & Veggie Bar	\$71,716.00	
You	\$71,174.00	
Vitafusion	\$69,507.00	
The Juice Caboose	\$68,957.00	
Healthy Dose	\$63,266.00	
Smoothie-Licious	\$61,174.00	
Just Juice	\$60,841.00	
Fresh Fusion	\$59,174.00	
Juice Do It	\$58,632.00	

Note. Final profit standings from the Juice Mogul: Oligopoly simulation (video.franklin.edu), showing a cumulative profit of \$71,174.00 and a fifth-place finish among the twelve competing firms.